

Solutions to Application-Based Cryptography & Security Questions

Course Assignment: Cryptography & Forensic Analysis Solutions

Scenario 1: Event Communication Security

Encryption Walkthrough (Caesar Cipher Shift of 4):

This security scenario utilizes a classical Caesar cipher where each character in the plaintext is shifted forward in the alphabet by 4 positions: $C = (P + 4) \bmod 26$

- **Plaintext:** MEET AT NOON
- **M (12) + 4 = Q (16):** M → Q
- **E (4) + 4 = I (8):** E → I
- **E (4) + 4 = I (8):** E → I
- **T (19) + 4 = X (23):** T → X
- **A (0) + 4 = E (4):** A → E
- **T (19) + 4 = X (23):** T → X
- **N (13) + 4 = R (17):** N → R
- **O (14) + 4 = S (18):** O → S
- **O (14) + 4 = S (18):** O → S
- **N (13) + 4 = R (17):** N → R

Encrypted Ciphertext: QIIX EX RSOS

Decryption Process for Authorized Members

An authorized recipient who possesses the shared secret key (Shift = 4) reverses the algorithm. For each letter in the received ciphertext, they shift backward by 4 positions: $P = (C - 4) \bmod 26$. Moving backward from Q by 4 positions restores the letter M, fully reconstructing the original message: **MEET AT NOON**.

Scenario 2: Intercepted Suspicious Message

Intercepted Ciphertext: GSRH RH Z HVXIVG

Deduction & Cryptanalysis Walkthrough

1. Single-Letter Word Identification: The ciphertext contains a single-letter word 'Z'. In the English language, the most common single-letter words are 'A' or 'I'. If 'Z' maps to 'A', it suggests the classical

Atbash Cipher, which is a monoalphabetic substitution cipher where the alphabet sequence is completely reversed ($A \leftrightarrow Z$, $B \leftrightarrow Y$, $C \leftrightarrow X$, and so on).

2. Key Mapping & Frequency Testing: Let's test the Atbash reverse-alphabet translation logic on the first word 'GSRH':

- G (7th letter from start) $\leftrightarrow$ T (7th letter from end)
- S (19th letter) $\leftrightarrow$ H (19th letter from end)
- R (18th letter) $\leftrightarrow$ I (18th letter from end)
- H (8th letter) $\leftrightarrow$ S (8th letter from end)

Word Decoded: **THIS**

3. Complete Message Translation:

- GSRH $\rightarrow$ **THIS**
- RH $\rightarrow$ **IS**
- Z $\rightarrow$ **A**
- HVXIVG $\rightarrow$ SECRET (H $\rightarrow$ S, V $\rightarrow$ E, X $\rightarrow$ C, I $\rightarrow$ R, V $\rightarrow$ E, G $\rightarrow$ T)

Decoded Plaintext Message: **THIS IS A SECRET**

Scenario 3: Secure Messaging System

Step-by-Step Vigenère Encryption Process:

Parameter	Char 1	Char 2	Char 3	Char 4	Char 5
Plaintext (P)	H (7)	E (4)	L (11)	L (11)	O (14)
Keyword (K)	K (10)	E (4)	Y (24)	K (10)	E (4)
Formula: $(P + K) \bmod 26$	$(7+10)=17$	$(4+4)=8$	$(11+24)=35 \rightarrow 9$	$(11+10)=21$	$(14+4)=18$
Ciphertext (C)	R	I	J	V	S

Influence of the Repeating Keyword

In polyalphabetic substitution ciphers like Vigenère, the repeating keyword creates a dynamic shift pattern. Because the key length (3 letters: 'KEY') is shorter than the plaintext length, the pattern repeats ('KEYKE').

Effect on Repeating Characters: Observe how the letter 'L' occurs sequentially at positions 3 and 4 in 'HELLO'. Because they align with different key letters ('Y' and 'K' respectively), they encrypt to completely different ciphertext letters ('J' and 'V'). This breaks the single-letter frequency distribution, greatly strengthening the encryption compared to a simple monoalphabetic substitution.

Scenario 4: Suspicious Digital Asset Investigation

Detection Techniques for Hidden Data (Steganalysis)

1. Statistical Steganalysis (Chi-Square Analysis):

Digital forensic tools analyze pixel color distributions within the logo. In unmodified images, color frequencies possess predictable patterns. Injecting payload data introduces visual noise or distinct byte alterations, altering the natural entropy of the file. Chi-Square statistical tests quantify these variations to detect the presence of hidden messages.

2. Bit-Plane Extraction (LSB Analysis):

Forensic investigators isolate specific bit planes, typically focusing on the Least Significant Bits (LSB). While standard image LSBs contain random, grain-like visual noise, an image hosting steganographic content often reveals artificial structured patterns or clear regions of higher density representing the embedded data stream.

Techniques to Strengthen Steganography Security

1. Pre-Encryption of the Payload:

Before embedding the payload into the digital asset, it must be encrypted using standard strong cryptographic algorithms (such as AES-256). This ensures that if forensic investigators successfully detect and extract the LSB payload, it remains completely unreadable and cryptographically secure.

2. Spread Spectrum Steganography:

Instead of sequentially embedding bytes into contiguous pixels (which creates an obvious, clustered modification footprint), spread spectrum techniques distribute the secret payload across pseudorandomly selected pixels and color channels using a shared cryptographic seed. This mimics natural image sensor noise, rendering visual and statistical detection highly improbable.

Scenario 5: Legacy Encryption in Corporate Communication

a. Reconstructing the Transposition Message

The received message is: **WECRLTEERDSOEFEAOCAIVDEN** (Length = 26 characters). This is encrypted using a three-level Rail Fence (transposition) cipher.

Step 1: Divide the rails based on wave calculations:

For a period of 4 (wave completes every 4 positions), the character counts per rail for 26 characters are:

- Rail 1 (Top): 7 characters → **W E C R L T E**
- Rail 2 (Middle): 12 characters → **E R D S O E E F E A O C**
- Rail 3 (Bottom): 7 characters → **A I V D E N**

Step 2: Reconstruct the zig-zag wave structure:

W . . . L . . . O . . . A . . . E . . .
. E . R . T . E . E . F . O . A . N . .
. . C . . . E . . . E . . . I . . . D .

By reading along the zig-zag path, the original plaintext is revealed:

Original Message: WE ARE DISCOVERED FLEE AT ONCE

b. Critical Evaluation of Vulnerabilities

Transposition ciphers only rearrange the order of letters; they do not alter the characters themselves. Consequently, they are extremely vulnerable to modern cryptanalysis due to:

1. Unaltered Frequency Distribution: The character frequencies in the ciphertext match those of normal English exactly. An attacker analyzing the file can easily identify that 'E', 'T', 'A' occur with high, natural frequency, instantly revealing a transposition cipher is in use.

2. Automated Anagramming: Modern computing systems can systematically brute-force millions of transposition combinations, geometric patterns, or grid alignments within fractions of a second to identify plain English words.

c. Proposed Practical Enhancement

To mitigate these structural vulnerabilities, the organization should transition to a **Product Cipher** framework, which sequentially combines both substitution and transposition methods. For example, first encrypt the plaintext message using a polyalphabetic key algorithm (such as AES or a Vigenère cipher), then transpose the resulting ciphertext using the Rail Fence structure. This effectively destroys both spatial sequence properties and character frequency signatures, providing robust modern confidentiality.